

PHILIP OLEMBO

Software Engineer | Full-Stack | Electrical & Electronics Engineer | Graphics Designer | Live Streamer
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EXPERIENCE SUMMARY

Full-stack software engineer with a background in Electrical and Electronics Engineering from Eastern Mediterranean University and several years of experience in live event production and graphic design. Completed a software engineering programme at Moringa School covering JavaScript, React, Python, Flask, SQL, and REST API development across five learning modules and a final project. Built and deployed multiple web applications, including an Employer-Centered Ideation (ECI) production-grade creative space booking platform with payment integrations, 360-degree space visualisation, and cloud asset delivery for Nairobi Streetwise.

KEY STRENGTHS

- **Full-Stack Development:** Built vision360, a full-stack creative space booking platform with a React and Node.js frontend and a PostgreSQL and Prisma backend, working in agile sprints alongside a parallel backend team and syncing daily on API contracts.
- **Independent Delivery:** Designed and shipped the MeterLink web app and its Python CLI companion independently, from scoping and architecture through to deployment, demonstrating the ability to own a product end-to-end.
- **API and Third-Party Integration:** Integrated M-Pesa Daraja STK push, PayPal, and bank transfer into vision360 alongside JWT authentication, Google OAuth, OTP email verification, and AWS S3 presigned asset delivery.
- **Teamwork and Collaboration:** Worked across solo and team engineering environments, leading the vision360 presentation site while collaborating across frontend and backend squads using Git workflows, pull requests, and code review.
- **Broadcast and Media Production:** Six years of live streaming and AV production experience across vMix, OBS, RTMP configuration, and multi-platform broadcast, providing strong intuition for real-time systems and user-facing reliability.

TECHNICAL SKILLS

Frontend: JavaScript (ES6+), React, HTML5, CSS3, React Router, Tailwind CSS, Vite, Three.js, Framer Motion, Recharts

Backend: Python, Flask, Node.js, Express, REST APIs, JWT Authentication, OAuth 2.0, Marshmallow, Nodemailer

Databases and ORMs: PostgreSQL, SQLite, SQLAlchemy, Prisma ORM, SQL (joins, subqueries, relations)

Tools and DevOps: Git, GitHub, VS Code, npm, pip, Jest, ESLint, Vitest, Vercel, Render

Cloud and Integrations: AWS S3, M-Pesa Daraja API, PayPal API, Google OAuth

Media and Design: Photoshop (Advanced), Illustrator, After Effects, Premiere Pro, InDesign, vMix, OBS

Engineering: AutoCAD, MATLAB, Multisim, Arduino IDE, Circuit troubleshooting, Microcontroller programming

PROJECT SUMMARIES

vision360 - Creative Space Booking Platform

vision-360-client-frontend.vercel.app

Stack: React 18, Vite, Three.js, Tailwind CSS, Radix UI, Framer Motion, Recharts, Node.js, Express, Prisma ORM, PostgreSQL, AWS S3, JWT, Google OAuth, M-Pesa Daraja, PayPal - Deployed: Vercel and Render

A full-stack platform that solves fragmented creative space booking by merging immersive 360-degree space discovery with an end-to-end booking, payment, and asset-delivery workflow. Targets photography clients, content creators, event planners, and property managers.

- Built the 360-degree panoramic space viewer using Three.js with drag-to-explore, inertia, auto-rotation, and multi-shot navigation.
- Implemented multi-method payments: M-Pesa STK push with async Daraja webhook confirmation, PayPal, and bank transfer.
- Integrated JWT and Google OAuth authentication and OTP email verification for secure user onboarding.
- Delivered secure post-session asset access using AWS S3 presigned URLs for one-click client downloads.
- Admin dashboard covers booking management, space CRUD, status updates, and an analytics view built with Recharts.
- Collaborated in agile sprints across parallel frontend and backend teams, syncing daily on API contracts via Git.

MeterLink - Prepaid Electricity Tracker (Web)

github.com/pijperfect/meterlink

Stack: *React*

A web-based platform that allows KPLC prepaid electricity customers to track their usage and token history parsed from SMS messages, providing visibility into consumption patterns.

- Built end-to-end as a solo project from concept through UI implementation and deployment.
- Designed for Kenyan consumers navigating the KPLC token-based electricity system.

MeterLink CLI - Prepaid Electricity Tracker (Python)

github.com/pijperfect/meterlink-cli

Stack: *Python, JSON, openpyxl*

A Python command-line companion to MeterLink that tracks prepaid electricity token purchases per meter using Excel import and local JSON storage.

- Implemented user authentication with hashed passwords, per-meter purchase history, running totals, and last-5-purchase summaries.
- Demonstrates Python OOP, file I/O, CLI design, and data persistence principles.

Motorized Camera Slider

EMU Final Year Project, January 2022

Stack: *Arduino (C++), NEMA 17 Stepper Motor, A4988 Motor Driver, Potentiometer, Limit Switch, MDF Wood, Copper Pipes, GT2 Timing Belt*

Team: *Amro Mohamedain, Philip Olembo, Canberk Balkir*

Supervised by: *Prof. Dr. Erhan Ince*

An Arduino-based motorised camera slider built to solve the challenge of manual camera tracking during online education in the COVID-19 pandemic. A 1-metre slider runs on copper pipe rails with a NEMA 17 stepper motor driving a GT2 timing belt, controlled by a potentiometer for variable speed and a limit switch for direction. Built from locally sourced materials for under \$200.

- Programmed the full motor control system in C++ using the Arduino Stepper library, mapping analogue potentiometer input to motor speed and using a limit switch to toggle forward and reverse stepper direction.
- Designed and fabricated the physical build from MDF wood and 22mm copper pipes, including the sliding platform, motor mount, electronics enclosure box, and pan/tilt camera mount.
- Applied engineering standards throughout: ISO 17396:2017, NEMA, EN310, IEC 60393, and IEEE std 1855.
- Presented to a jury of 6 professors and awarded 2nd Place at Eastern Mediterranean University's January 2022 Final Year Project Competition.

RELEVANT WORK EXPERIENCE

ECI Project | vision360 - Moringa Capstone Project (Team of 6)

May 2026

- Delivered a production-grade full-stack platform across parallel frontend and backend squads, completing the project within a 5-month timeline with a live deployment on Vercel and Render.
- Built the 360-degree space viewer using Three.js with drag-to-explore, inertia, auto-rotation, and multi-shot navigation.
- Engineered a 3-tier booking flow (Essential, Professional, Enterprise) connected to the backend REST API, handling date selection, space type, and special requirements.
- Integrated 3 payment methods and full authentication coverage including JWT, Google OAuth, and OTP email verification.
- Implemented secure asset delivery via AWS S3 presigned URLs and built the analytics dashboard using Recharts.
- Managed Git-based collaboration with pull requests across 4 agile sprint phases and daily API contract syncs.

Media and Streaming Coordinator | Freelance / Church and Events

Jan 2019 - Present

- Set up and operated live streaming productions using vMix and OBS for church services, weddings, conferences, and outreach events across Facebook, YouTube, TikTok, and Instagram simultaneously.
- Configured RTMP stream keys, stream URLs, and multi-platform distribution pipelines, maintaining uptime and stream quality under live production conditions.
- Managed sound engineering with analogue mixers, coordinating technical setup, rehearsal, and real-time troubleshooting.
- Produced branding assets including logos, animated intros, lyrical video overlays, and event flyers for businesses and organisations.

Intern | Bar-er Energy Ltd.

Jan 2023 (40 days) @ Famagusta, TRNC

Electrical and Electronics Engineering / Renewable Energy

- Completed electrical project design using AutoCAD for solar panel installation layouts and wiring schematics.
- Gained hands-on experience in solar panel installation, wiring troubleshooting, preventive maintenance, and safety protocol adherence.
- Bridged theoretical engineering studies with real-world renewable energy systems.

EDUCATION

Software Engineering Bootcamp | Moringa School (Flatiron Curriculum)

Nov 2025 - May 2026

Intensive 6-month full-stack programme across 5 structured modules. Built and deployed multiple solo and team projects throughout.

Module 1 - Foundations: HTML, CSS, JavaScript fundamentals, Git and GitHub, CLI, algorithms, arrays, objects, debugging, SDLC, VS Code, Node.js

Module 2 - JavaScript: Functions, scope, higher-order functions, callbacks, DOM manipulation, events, HTML forms, async JS, Promises, async/await, external APIs, fetch, unit testing (Jest), ESLint, Big O notation

Module 3 - React: Components, props, state (useState), event handlers, controlled forms, useEffect, useRef, useContext, custom hooks, CRUD, data fetching, React Router, Vitest, npm

Module 4 - Python and OOP: Python fundamentals, lists, tuples, dictionaries, sets, loops, functions, PyPi, pip, file I/O, OOP (classes, inheritance, property decorators), one-to-many and many-to-many relationships, CLI design, debugging and testing

Module 5 - Flask and Backend: Flask, REST APIs, full CRUD, Flask-SQLAlchemy, SQL (SELECT, filter, JOIN, subqueries), Marshmallow serialization, JWT authentication, sessions, cookies, password hashing, deployment (Render), data structures and algorithms (linked lists, hash tables, trees, stacks, queues, recursion), React-Flask integration

B.Sc. Electrical and Electronics Engineering (Honours) | Eastern Mediterranean University

2018 - 2022

Graduated with CGPA 3.14 / 4.00. Specialised in circuit design, microcontroller programming, renewable energy systems, and engineering simulation (AutoCAD, MATLAB, Multisim). Final Year Project: Motorized Camera Slider - 2nd Place.

ADDITIONAL INFORMATION

Languages:

English (fluent), Swahili

Volunteer and Community:

Provided live streaming and Audio-Visual technical support for church community events, conferences, and outreach programmes since 2019, including coordinating multi-camera setups and real-time social media broadcasts.

PERSONAL INTERESTS

Building tools that solve real-world African infrastructure challenges (energy, payments, logistics). Live event production and streaming technology. Graphic design and motion graphics. Exploring the intersection of software engineering with creative media.

REFEREES

Daniel Karanja

Technical Mentor | Moringa School, Nairobi
University

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Prof. Dr. Erhan Ince

Supervisor and Student Advisor | Eastern Mediterranean

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